Method

We applied the group's NSGA-II implementation to ZDT3 and vehicle crashworthiness, then used constrained NSGA-II on MW7 and RCM20. Selection combines non-dominated rank and crowding distance; simulated binary crossover and polynomial mutation generate offspring. For the constrained problems, feasible points outrank infeasible points, and infeasible points are ordered by total constraint violation.

Task 1. ZDT3

The 30-variable benchmark has a disconnected true front [1]. With a population of 100 and max_gen=500, the final objective-space scatter covers all five separated segments (Fig. 1). Crowding distance favours sparsely sampled regions, which can help retain coverage across the disconnected front.

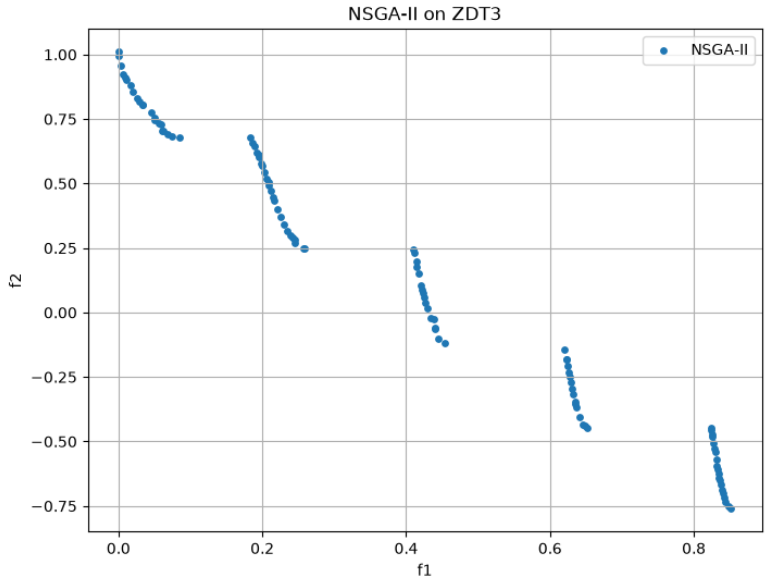


Fig. 1. ZDT3 final population from the group's NSGA-II run.

Task 2. Vehicle crashworthiness

We solved the five-variable crashworthiness model [2] with a population of 100, max_gen=500 and seed 2026. NSGA-II returned 100 designs (98 distinct). Weight, acceleration and toe-board intrusion trade off: designs with lower intrusion can have greater weight or acceleration. The pairwise views show clusters and sparsely sampled gaps (Fig. 2). The nonlinear mapping from five thickness variables to three objectives and crowding distance acting only among sampled solutions can both contribute to uneven spacing. Larger populations or stronger mutation should be compared across seeds at similar evaluation budgets.

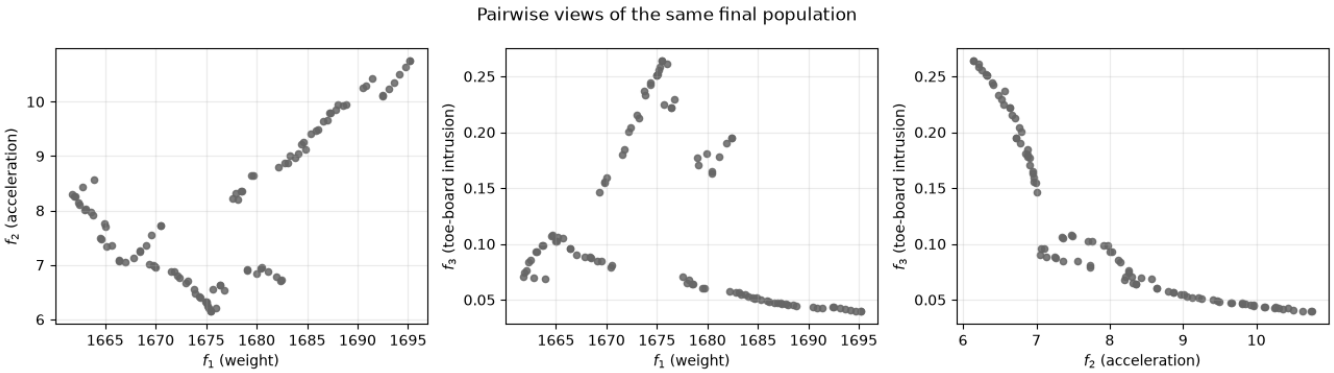


Fig. 2. Pairwise projections of the crashworthiness final population (100 designs).

Task 3. Constrained NSGA-II on MW7

For MW7 [3], five seeds (0-4) used N=100, 500 evolutionary updates, crossover probability 0.8 and per-variable mutation probability 0.05. All final populations were fully feasible. Seeds 0, 3 and 4 covered all three visible branches; seeds 1 and 2 stayed on the upper-left branch (Fig. 3). Feasibility-first selection removes infeasible candidates, while variation around selected parents can keep the search near reached branches. Crowding distance spreads points within a branch but cannot discover a missing one. Larger populations or stronger mutation should be compared across seeds at similar evaluation budgets to assess branch coverage.

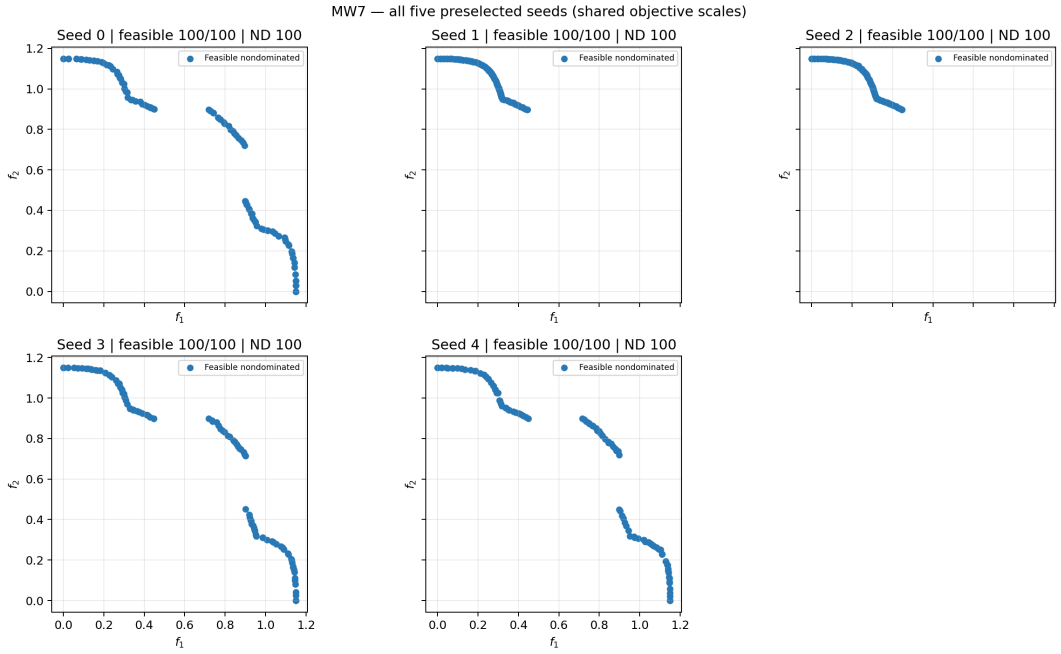


Fig. 3. MW7 final populations for seeds 0-4 on shared objective scales.

Task 4. RCM20 model and experiment

RCM20 has four variables, two objectives and seven inequality constraints [4]. We implemented the authors' MATLAB code in *CEC2021_func.m* (func == 20) [5]. The authors' supplementary material differs from this code in the formula for P, the pressure constraint (P0 ≥ 1000 in the supplement versus P0 ≤ 1000 in the code), and the temperature constraint, which uses P0 instead of ΔT in the supplement. Within the stated bounds, the printed formulas give negative P0 wherever defined, conflicting with P0 ≥ 1000. Following the TA's advice, we therefore used the repository implementation, which admits feasible solutions. Constraint violation is the sum of positive constraint violations. One-factor sweeps varied population size N, generation limit G, crossover probability pc and mutation probability pm over five seeds. A second stage compared N-G splits near 200,000 evaluations; a third stage combined the best levels at exactly 200,000 evaluations. The setting selected by median HV with a diversity tie-break was tested on ten new seeds (100-109).

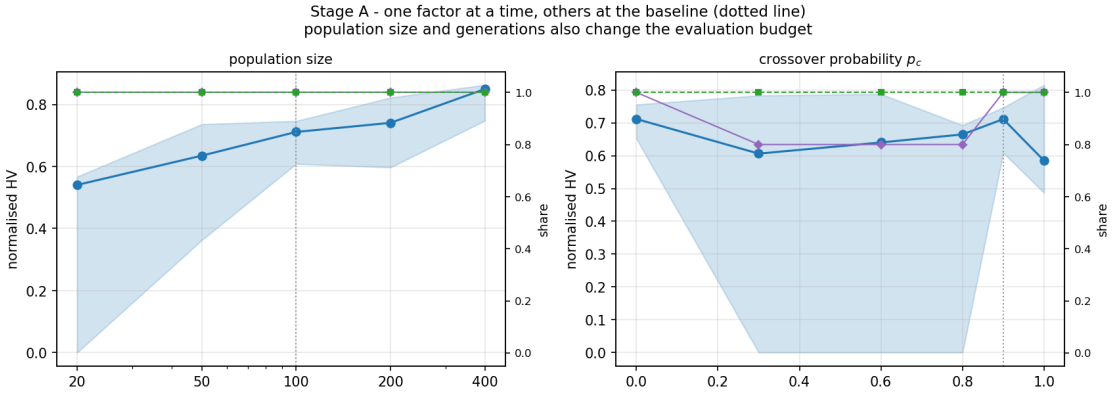


Fig. 4. RCM20 Stage A upper panels: N and pc (five seeds).

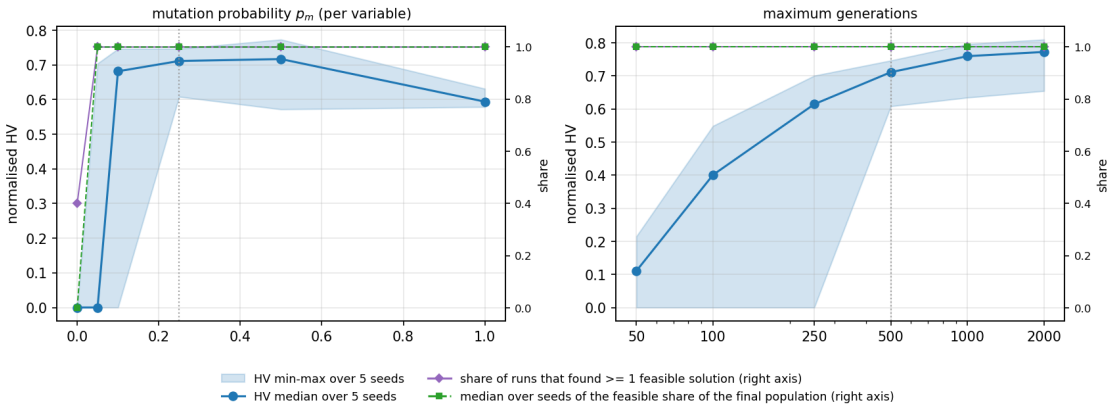


Fig. 4 (continued). Lower panels: pm and G; legend applies to all four panels.

Task 4. RCM20 results

Mutation had the largest observed effect: with pm=0, three of five runs found no feasible solution, while pm=0.1-0.5 yielded median normalized hypervolume (HV) of 0.68-0.72. Increasing N or G improved HV in the one-factor sweeps but also increased evaluations. Near 200,000 evaluations, different N-G splits gave median HV of 0.76-0.85. Crossover effects varied with N and pm.

Configuration	Feasible runs	Median HV	Median distinct
Selected: N=400, G=499	10/10	0.839	261.5
Equal-budget baseline: N=100, G=1999	9/10	0.740	86.0

Table 1. Held-out seeds 100-109. Both settings use 200,000 objective evaluations; pc=0.9 and pm=0.25.

The selected configuration had higher HV on seven of ten paired seeds and produced more distinct solutions, although its median largest spacing gap was also larger (0.136 versus 0.051). Pooling its ten runs and removing duplicates yielded 598 feasible, mutually non-dominated designs (Fig. 5). A known feasible reference point dominates the obtained lowest-f1 extreme, indicating room for further convergence at that end.

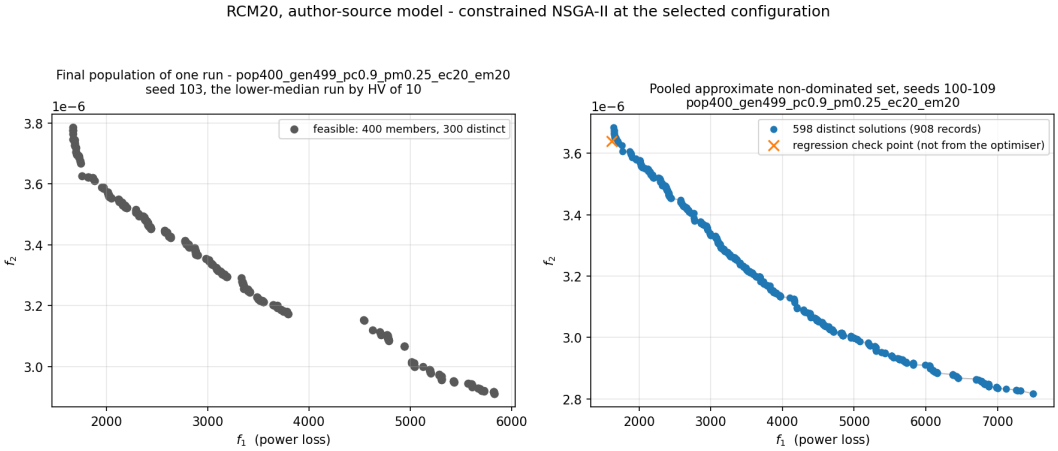


Fig. 5. RCM20 final population of seed 103 (left) and 598 distinct pooled solutions from ten seeds (right); cross: known feasible point.

References

[1] E. Zitzler, K. Deb and L. Thiele, *Evolutionary Computation* 8(2), 173-195 (2000).

[2] X. Liao et al., *Structural and Multidisciplinary Optimization* 35, 561-569 (2008).

[3] Z. Ma and Y. Wang, *IEEE Transactions on Evolutionary Computation* 23(6), 972-986 (2019).

[4] A. Kumar et al., *Swarm and Evolutionary Computation* 67, 100961 (2021).

[5] P. N. Suganthan, 2021-RW-MOP GitHub repository, *CEC2021_func.m*, github.com/P-N-Suganthan/2021-RW-MOP (accessed 23 Oct 2026).